

Physiology: Transport of Urea, Glucose & Organic Solutes

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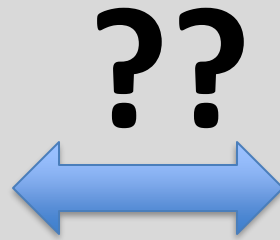
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You just started your rotation in Emergency Medicine when a 12-year-old girl is brought in to the ER by her parents. She is an avid soccer player and has no significant PMH. She was 167 cm tall, and her weight was 48 kg (BMI = 17 kg/m²). Her parents are very concerned because she's lost a lot of weight in the past month and is feeling lethargic and is easily fatigued. They state that her schoolwork is suffering.

Her history is significant for frequent bathroom trip during the day (polyuria) and she complained of being thirsty all the time. Her PE revealed signs of dehydration. Diuresis was quantified as very high (7.5L/day) and urinalysis showed proteinuria (+++), and glycosuria (+++) with normal glycemia.

You start thinking how these are all related.....

**Fatigue
Lethargy**



**Frequent Urination
Thirst
Sugar in Urine**

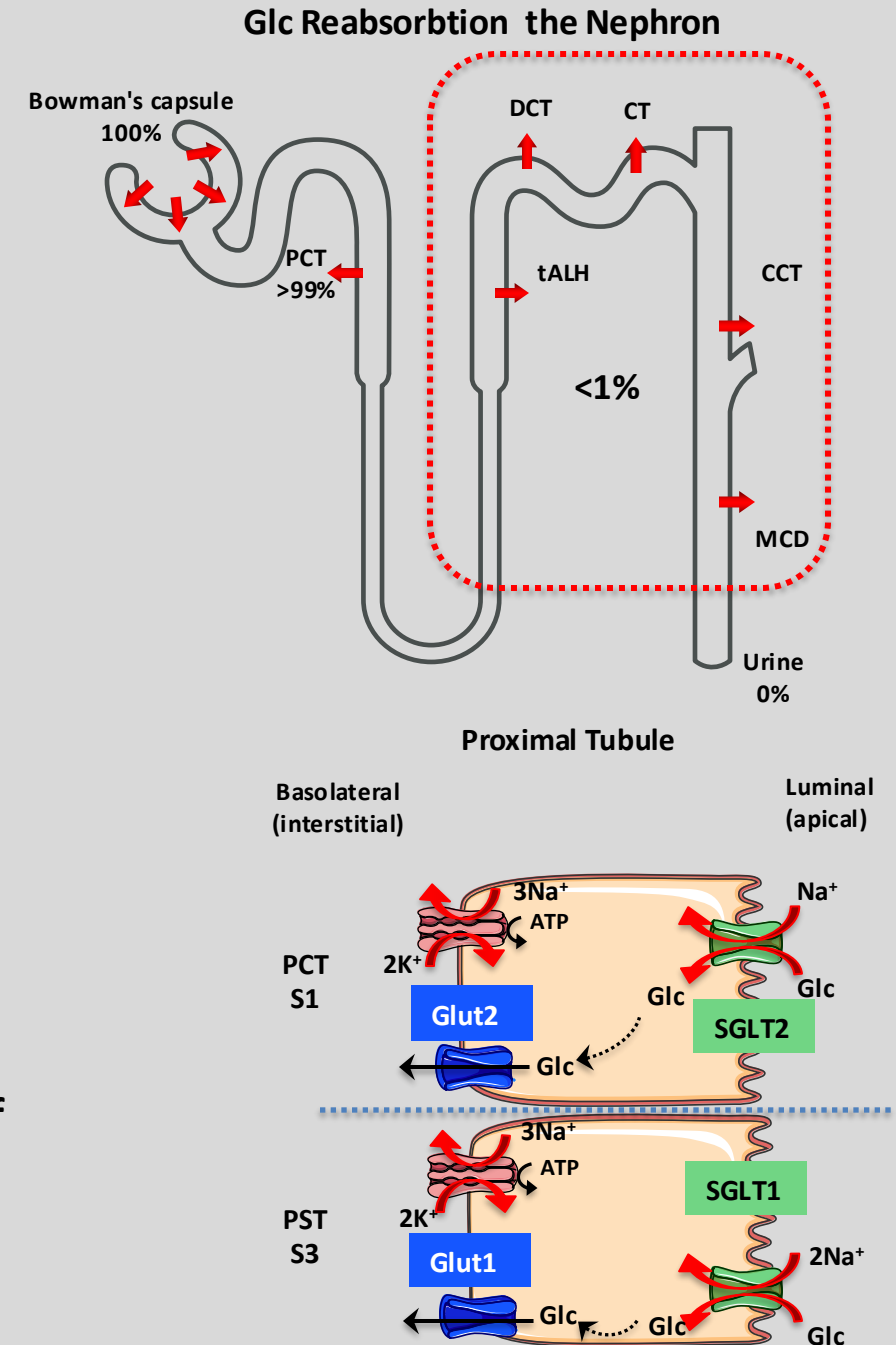
Session Objectives

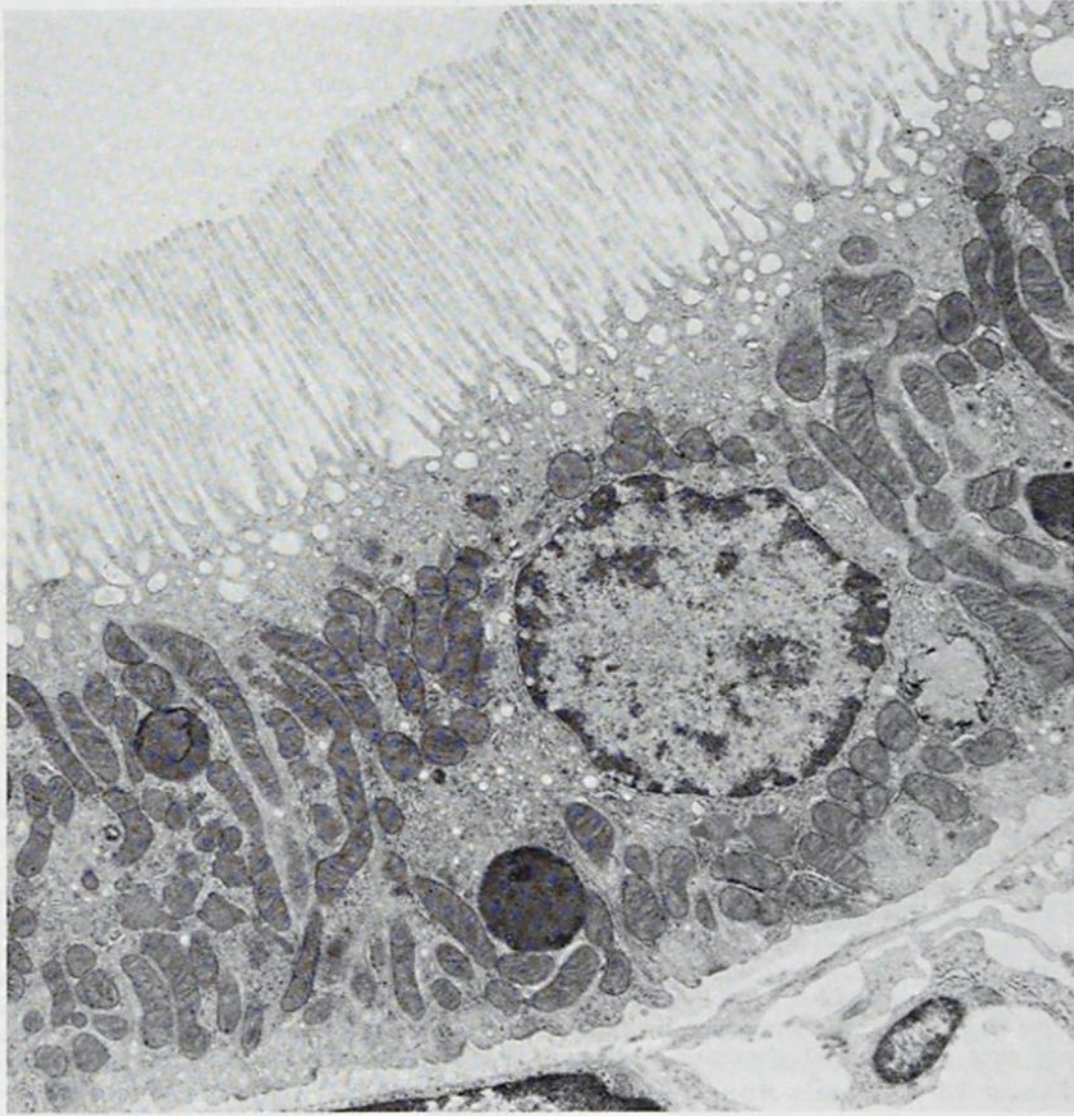
At the conclusion of the session, the student should be able:

- 1. Define & describe the mechanisms underlying the reabsorption of glucose, urea & urate**
- 2. Define & describe the mechanisms underlying the reabsorption of proteins, oligopeptides & amino acids.**
- 3. Define & describe the mechanisms underlying the secretion of organic acids and bases.**
- 4. Explain the significance of the organic acid and base secretory systems.**

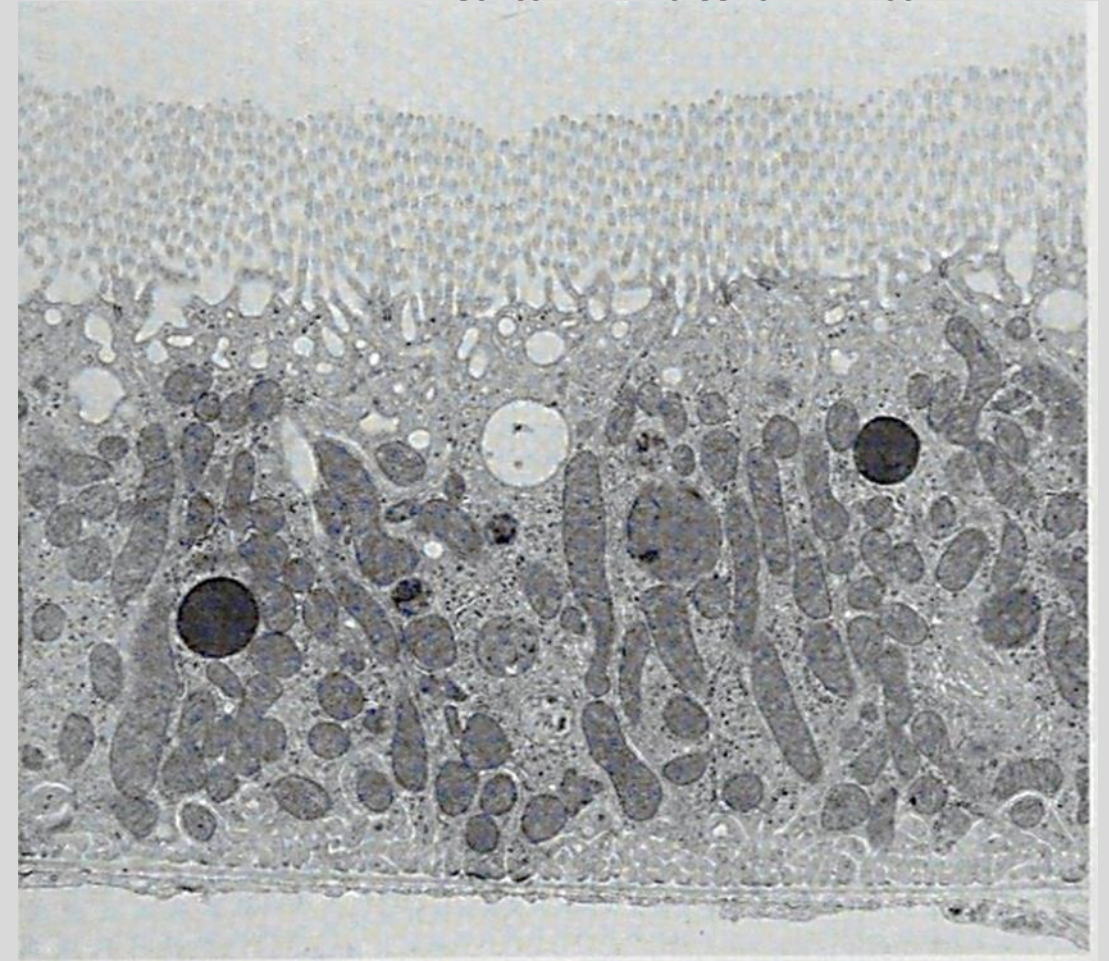
Renal Regulation of Glucose

- Plasma glucose $[Glc]_{PI}$:
 - Remains within a narrow range (fasting $Glc \cong 4\text{-}5\text{mM}$ (70-100 mg/dL)).
 - Regulated by insulin & other hormones.
 - Freely filtered at the glomerulus
 - Completely reabsorbed in the PT (transcellularly).
- PCT (S1 segment):
 - Na^+ -Glc cotransporter 2 (SGLT2):
 - Apical transporter
 - High-capacity, low-affinity.
 - Responsible for 90% of Glc reabsorption.
- PST (S3 segment):
 - Na^+ -Glc cotransporter 1 (SGLT1):
 - Low-capacity, high-affinity,
 - Has 2:1 $Na^+ \rightarrow Glc$ stoichiometry
 - Can generate a far larger glucose gradient across the apical membrane.
- SGLT2 inhibitors are novel “glucuretics” that have been approved for the treatment of T2DM & CHF.
- Glc exits across the basolateral membrane via GLUT (Glut1 & Glut2) family of GLC channels (distinct from SGLT)
- They are Na^+ -independent and move Glc via facilitated diffusion.





22. Transmission electron micrograph of S_1 segment of rat proximal tubule. The cells are characterized by a tall brush border, a prominent nucleus, and a lysosomal apparatus, and extensive invaginations of the basolateral plasma membrane. (Magnification $\times 10,600$.)



23. Transmission electron micrograph of S_2 segment of rat proximal tubule. The brush border is less prominent than in the S_1 segment, and the cell is characterized by a nucleus, a lysosomal apparatus, and extensive invaginations of the basolateral plasma membrane. (Magnification $\times 10,600$.)

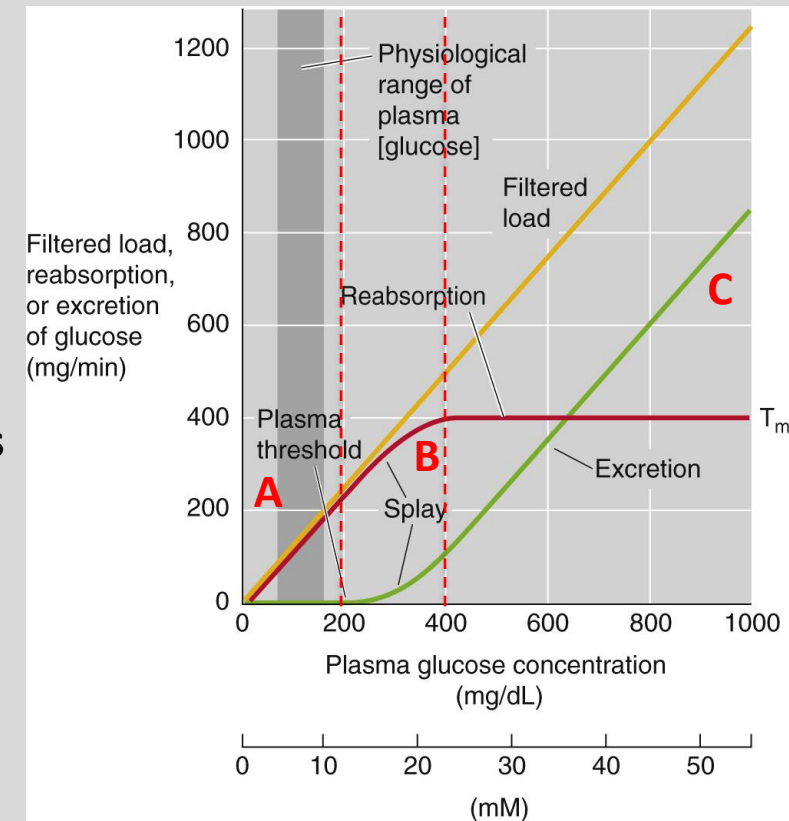
Urinary Excretion of Glucose

- Glucose titration curve: The relationship between $[Glc]_{PI}$ & rate of glucose reabsorption via glucose infusion (*iv*).
- Low FL glucose (< 200 mg/dL) $\rightarrow$ no glucose in the urine $\rightarrow Glc_{CI} = 0$. (A)
- Above a threshold of ~ 200 mg/dL (~ 11 mM) Glc appears in the urine (glucosuria).
- As FL increases $\rightarrow \uparrow Glc_{Ex}$ (linear) $\rightarrow \uparrow Glc_{CI}$ (B)
- At high Glc loads (C), amount of Glc reabsorbed becomes small compared with the filtered load $\rightarrow$ Glc remains in the tubular lumen.
- Glc_{Exc} rises as Glc_{PI} increases.
- Threshold (~ 200 mg/dL) is higher than the normal $[Glc]_{PI}$ of ~ 100 mg/dL (~ 5.5 mM), and because the body effectively regulates $[Glc]_{PI}$
- $\therefore$ healthy people do not excrete any glucose in the urine, even after a meal.
- Even in people with DM, glucosuria doesn't occur until the blood sugar level exceeds threshold.
- Rate of glucose reabsorption ~ 400 mg/m $\rightarrow$ plateau $\rightarrow$ Transport maximum (T_m).
- T_m : SGLTs are fully saturated $\rightarrow$ Transporters can't respond to increases in Glc F.L.
- Splay: When rate of Glc reabsorption reaches the T_m gradually.
- Reflects both anatomical and kinetic differences among nephrons.
- $\therefore$ a nephron's Glc F.L. may be mismatched to its capacity to reabsorb glucose (e.g., a nephron with a larger glomerulus has a larger load of glucose to reabsorb, different nephrons may have different distributions and densities of SGLT2 and SGLT1 along the PT).

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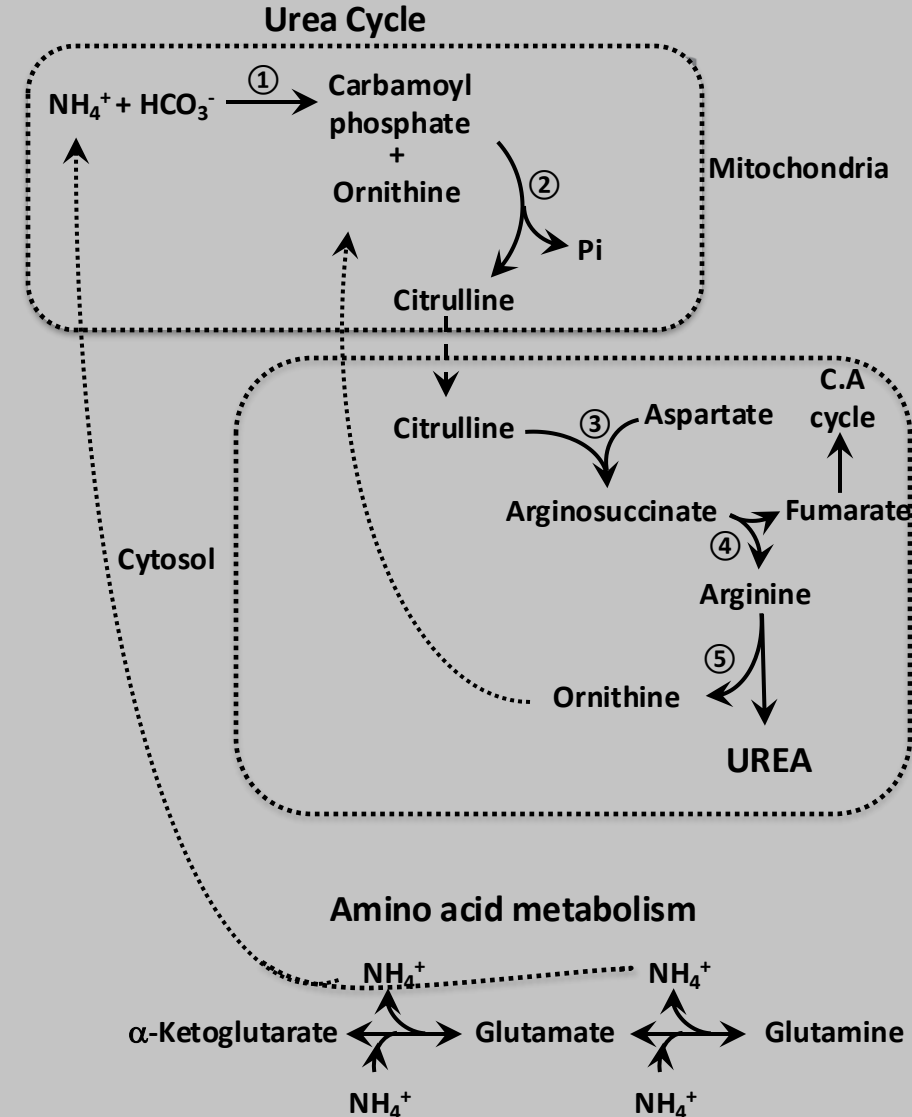
Glc Titration Curve



Renal Regulation of Urea

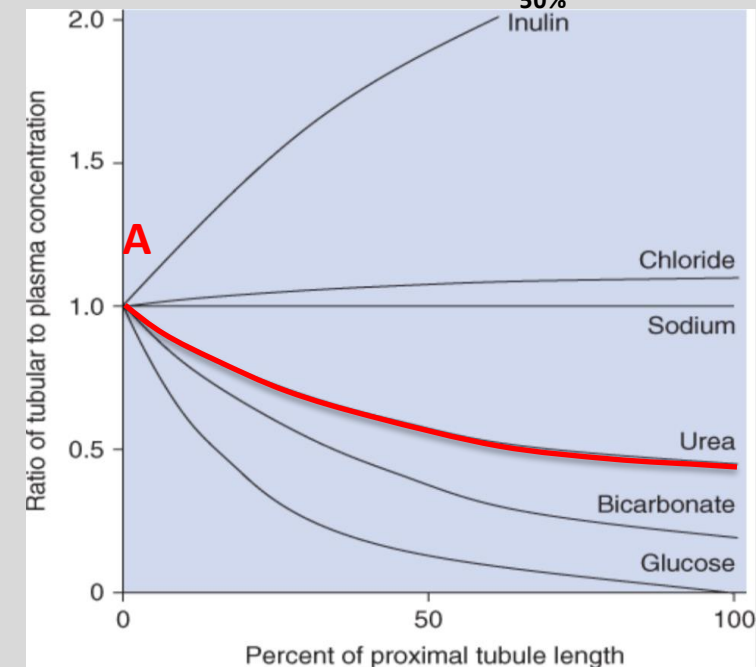
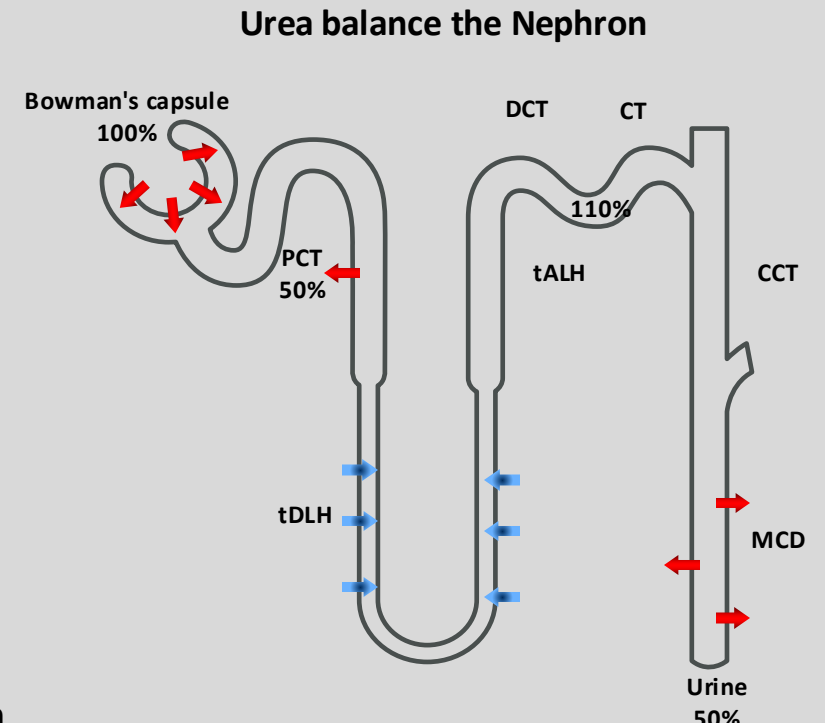
- Urea cycle: Responsible for the removal of excess N and NH_4^+ (ammonium).
 - Detoxifies ~95% of the NH_4^+ (liver).
 - Generates urea from NH_4^+ , the primary nitrogenous end product of amino-acid catabolism.
- Urea (Ur) is a very special substance for the kidney.
 - MW: 60 Da, highly polar & H_2O soluble.
 - Normal plasma concentration of urea: 2.5 to 6 mM.
 - End product of protein metabolism, a waste product to be excreted, & critical for H_2O regulation and salt reabsorption.
 - Provides a critical means of ridding the body of N waste.
- Primary route of excretion: urine (some urea exit via stool & sweat).
- Freely filtered at the glomerulus, and then reabsorbed and secreted.
- Tubules reabsorb more urea than they secrete
- $\therefore$ amount of urea excreted in the urine < amount filtered.
- Normal levels of urea are not toxic
- Large amounts are produced in a high-protein diet, representing a large osmotic load that must be excreted.
- Clinical laboratories report plasma urea levels as blood urea nitrogen (BUN)
- Normal BUN values: 7 to 18 mg/dL.

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Urea Transport in the Nephron

- Urea is not bound to plasma proteins.
- Constitutes about half of the solute content of the urine.
- Does not permeate lipid bilayers.
- Urea is both reabsorbed & secreted in the nephron.
- The tubules reabsorb more urea than they secrete
- Amount of urea excreted in the urine $\ll$ Quantity filtered (~50%)
 - Reabsorption: Proximal tubule & medullary collecting duct
 - Secretion: Thin descending & ascending limbs of the loop of Henle.
- In the very early proximal tubule, $[Ur]_L = [Ur]_{Pl}$ (A)
- H_2O reabsorption increases $[Ur]_L$ in the lumen, a favorable transepithelial urea gradient drives urea reabsorption.
- Urea transport depends $\Delta[Ur]$ across the tubule epithelium
- Changes in urine flow affect renal urea handling (e.g., urea excretion rises with increasing urinary flow).
- At low urine flow, tubule reabsorbs more H_2O & $\therefore$ kidneys excrete ~15% of filtered urea.
- At high urine flow $\rightarrow$ tubules reabsorb less H_2O & urea $\rightarrow$ kidneys may excrete as much as 70% of filtered urea.
- Progressive renal disease $\rightarrow$ dec GFR $\rightarrow$ low urine flow $\rightarrow$ urea retention $\rightarrow \uparrow$ BUN



Urea Transport in the Nephron

PT:

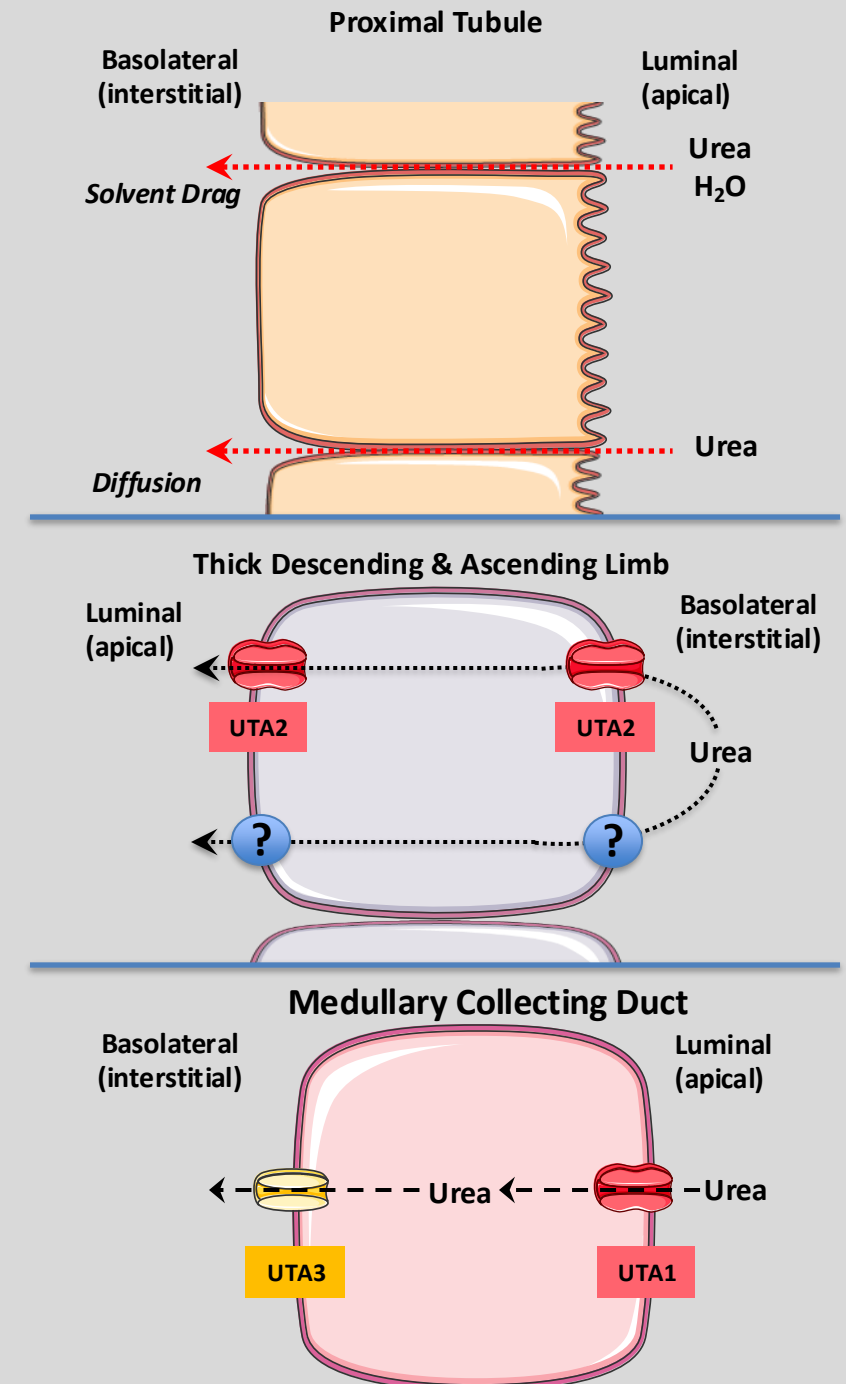
- ~ 50% of the FL of urea is reabsorbed **PARACELLULARLY**.
- Reabsorption occurs by solvent drag & diffusion across tight junctions.
- The greater the fluid reabsorption along the PT, the greater the reabsorption of urea.

Thin DL & AL:

- Amount equal to that reabsorbed is secreted back into the lumen of the loops of Henle (**TRANSCELLULAR**).
- This restores $[Ur]_L$ content to its original value.
- Secretion is via facilitated diffusion using urea transporters (UTs)
- UTs belong to the SLC14 family of solute transporters
- UTs exhibit saturation kinetics and other hallmarks of carrier-mediated transport.

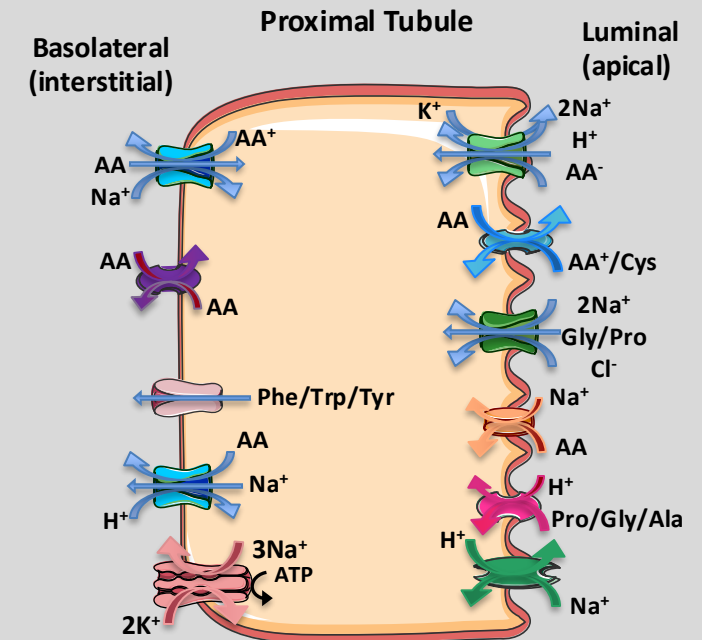
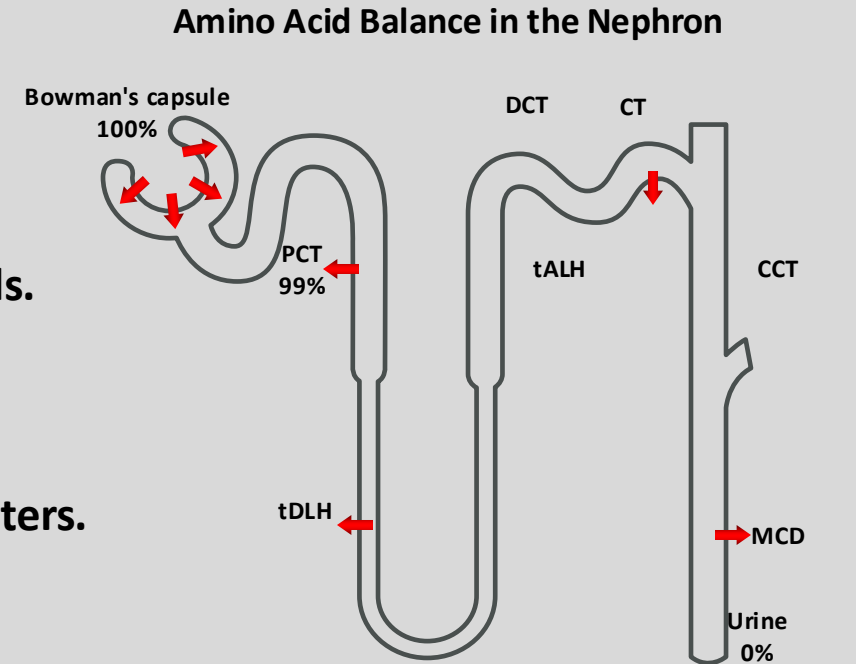
MCD:

- ~ 50% of the is $[Ur]_L$ reabsorbed (**TRANSCELLULAR**).
- Net result: 50% of FL of urea is ultimately excreted (assuming normal urine flow & hydration status).



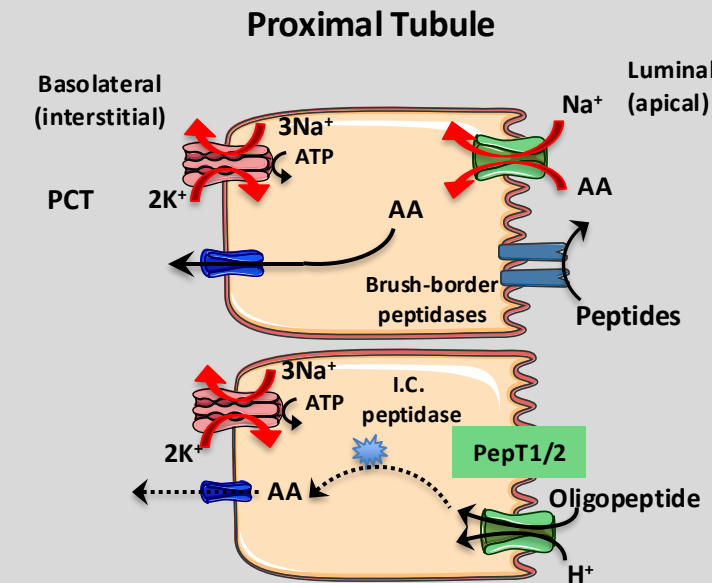
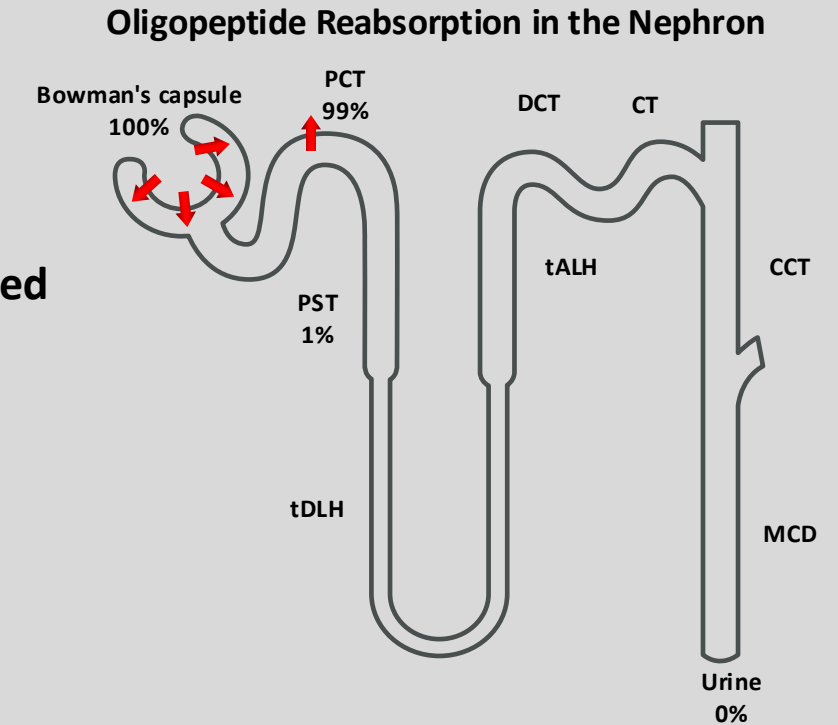
Amino Acid Transport in the Nephron

- Total concentration of amino acids (AA) in the blood is ~ 2.4 mM.
- Largely absorbed by the GI tract.
- Products of protein catabolism or *de novo* synthesis of nonessential amino acids.
- Freely filtered in the glomerulus.
- Because AA are important nutrients, they must be retrieved from the filtrate.
- PT reabsorbs $>98\%$ of AA via TRANSCELLULAR routes
- Use a wide variety of transporters of the Solute Carrier (SLC) family of transporters.
- SLCs do not have substrate specificity.
- Luminal (apical) membrane:
 - AA enters the cell via Na^+ - or H^+ -driven transporters
 - AA enter using AA exchangers.
- Basolateral (interstitial) membrane:
 - AA exit the cell via AA exchangers (some are Na^+ -dependent)
 - Na^+ -dependent uptake occurs in late PT (S3), where availability of luminal AA is low.
 - SLC38A3 mediates the transfer of AA important for cellular nutrition/metabolism across the basolateral membrane.



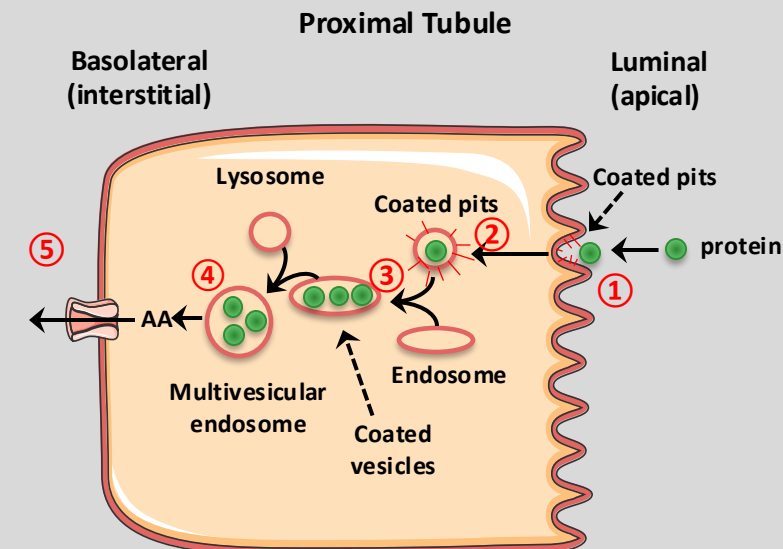
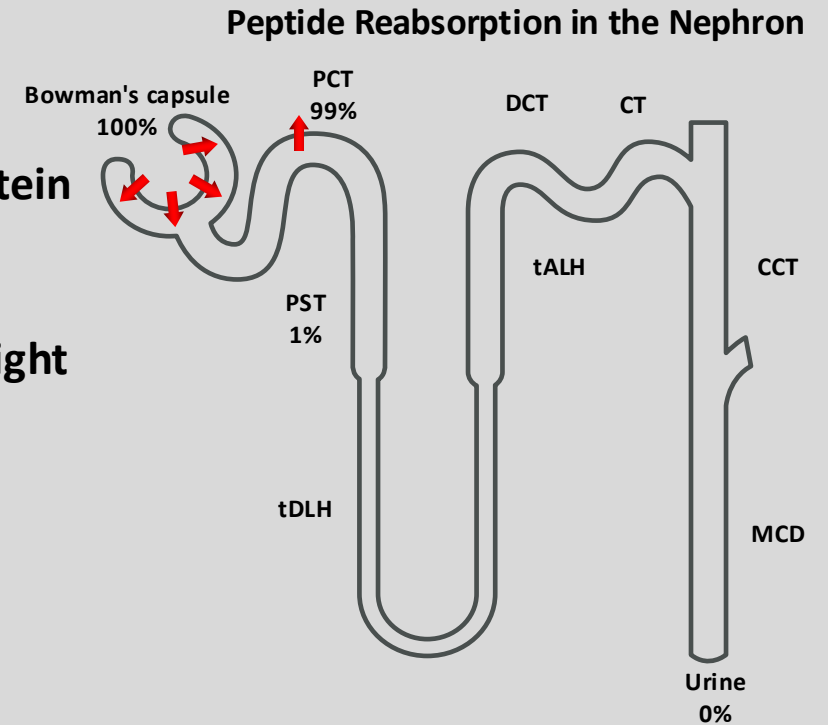
Oligopeptide Transport in the Nephron

- Oligopeptides (AA_n):
 - Short chains of AA (2-20 residues).
 - Unlike larger polypeptides or proteins, they are compact & serve specialized biological functions (e.g., signaling, immune responses, endocrine regulation).
- PT reabsorbs ~99% of filtered AA_n .
- Segments beyond the PT contribute little to peptide transport.
- Several peptidases are present at the outer surface of the brush-border membrane of PT cells (similar to GI tract)
- Brush-border enzymes:
 - Variety of enzymes hydrolyzing many peptides (e.g., γ -glutamyltransferase, aminopeptidases, endo- & dipeptidases)
 - Release AA into the tubule lumen
- Some AA_n (2-5 AA) are resistant to brush-border enzymes
- Enter cells via apical H^+/AA_n cotransporters PepT1/T2.
- PepT1: Low-affinity, high-capacity transporter in the PCT (S1-S2)
- PepT2: High-affinity, low-capacity transporter in the PST (S3).
- Intracellularly, AA_n are hydrolyzed by cytosolic peptidases
- Distinction between which AA_n get digested in the lumen vs I.C is not clear-cut.



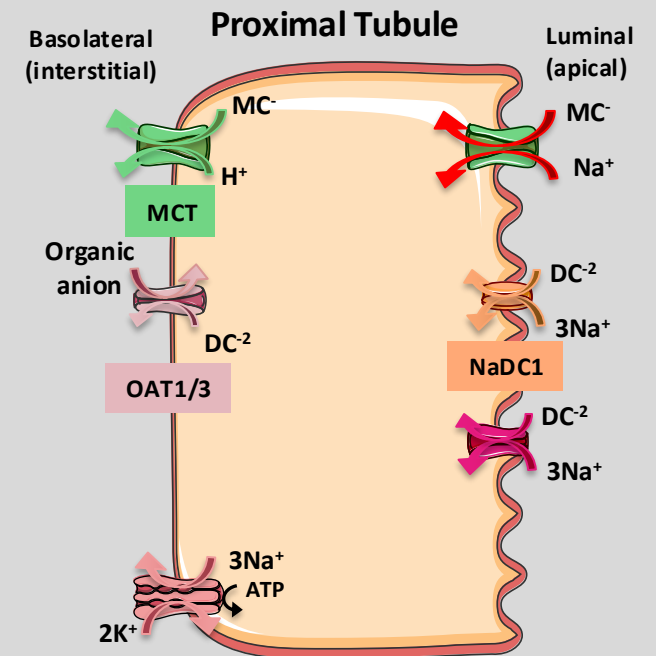
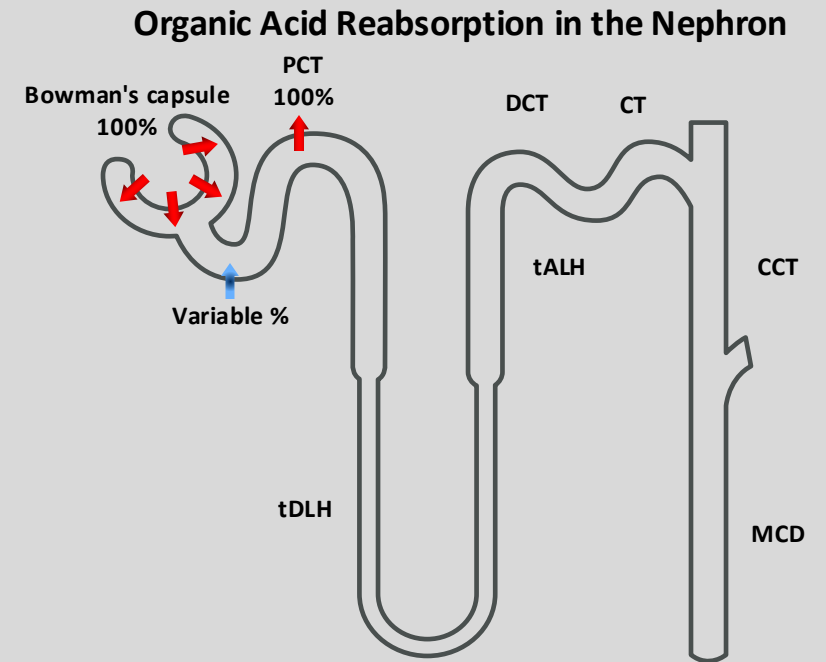
Protein Transport in the Nephron

- The glomerular filtration barrier prevents the filtration of large amounts of protein
- This restriction is incomplete (e.g., $[Alb]_{\text{filtrate}}$ is only 0.01-0.05% $[Alb]_{\text{pl}}$)
- Reabsorption of 96-99% of filtered albumin.
- Tubules also reabsorb relatively freely-filtered, LMW proteins (e.g., lysozyme, light chain Ig, β_2 -microglobulins) & polypeptide hormones.
- $\therefore$ tubule injury $\rightarrow$ proteinuria (even w/o glomerular injury).
- PT use receptor-mediated endocytosis to reabsorb proteins and polypeptides.
- First step ① is binding to a receptor complex at the apical membrane.
- Followed by internalization into coated endocytic vesicles ②.
- Factors that interfere with vesicle formation/internalization (e.g., metabolic inhibitors, cytochalasin-B).
- Vesicles fuse with endosomes ③; this fusion targets the vesicle for delivery to lysosomes.
- At the lysosomes, proteases digest the contents ④.
- PTs release LMW products of digestion across the basolateral membrane into the peritubular circulation ⑤.
- Hardly any reabsorption of intact proteins.
- Small subset of proteins avoids the lysosomes and moves by transcytosis for release at the basolateral membrane.



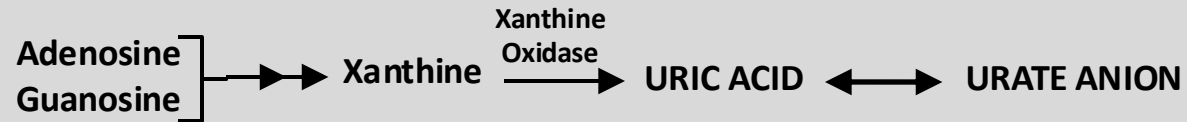
Organic Acid Secretory System (OASS)

- Mono- (R-COO^-), di- (R-2COO^-) & tricarboxylates (R-3COO^-): Products of Glc & fatty acid metabolism (anaerobic/citric acid cycle).
- Important for energy metabolism $\therefore$ urine loss is wasteful.
- PT reabsorbs virtually all these substances.
- $\uparrow [\text{Carboxylates}]_{\text{PI}} \rightarrow [\text{Carboxylates}]_{\text{U}}$
- Urinary excretion occurs when the FL of ketone bodies (acetoacetate, β -OH butyrate) exceeds their T_m in PT.
- $[\text{Carboxylates}]_{\text{PI}}$: 1-3 mM (lactate is the largest fraction).
- Na^+ -dependent cotransporters carry carboxylates across the apical membranes
- All are members of the Solute Carrier (SLC) family of cotransporters.
- R-COO^- exit across the basolateral membrane via MCT2.
- Efflux is coupled to $\text{H}^+ \rightarrow [\text{Na}^+]_{\text{IC}} \ \& \ \uparrow \text{pH}_{\text{IC}}$.
- R-2COO^- exit the cell across the basolateral membrane via the renal organic anion transporters (OAT1 & OAT3).
- OATs overlap in substrate specificity.
- Para-aminohippurate (PAH): Small (MW: 194 Da), H_2O -soluble organic anion.
- Freely filtered & secreted by the PT $\rightarrow$ More is removed than filtered.
- At low plasma concentrations, PAH entering the kidney is removed from the plasma and excreted.
- PAH clearance can be used experimentally as a measure of renal plasma flow (effective renal plasma flow).



Renal Regulation of Uric Acid

- Urate (Ur): Monovalent anion & the end product of purine catabolism



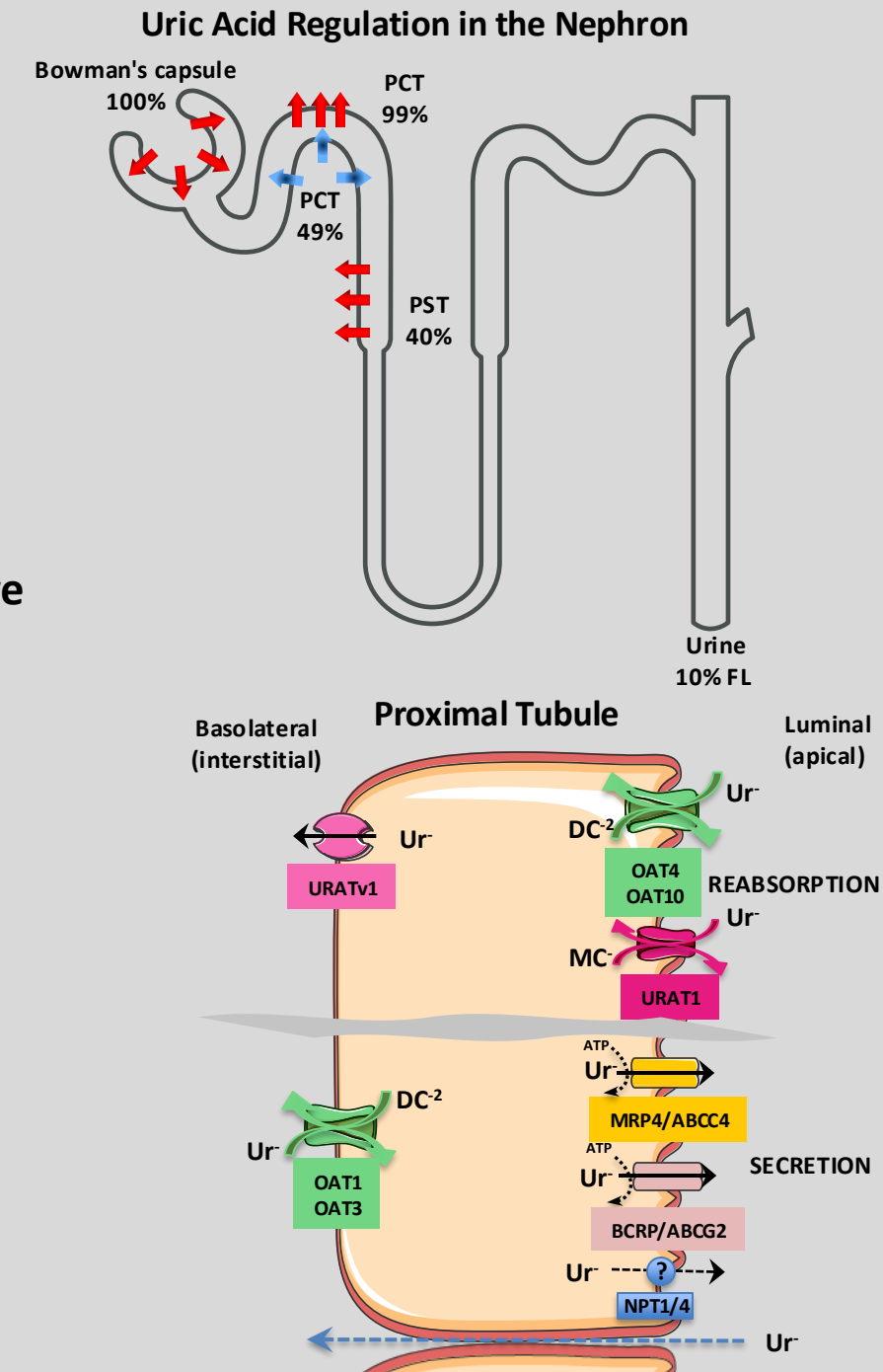
- $[\text{Ur}^-]_{\text{PI}} \cong 3\text{-}7 \text{ mg/dL (0.2-0.4 mM)}$
- Filtered, reabsorbed & secreted (in PT).
- Reabsorption is the more important transport pathway.

Reabsorption

- PT reabsorbs Ur^- via paracellular (passive diffusion) & transcellular routes (active transport).
- Paracellular reabsorption:
 - Important during EC volume depletion
 - Results in compensatory enhancement of PT fluid reabsorption.
 - $\therefore \uparrow [\text{Ur}^-]_{\text{L}} \rightarrow \uparrow \text{Paracellular } \text{Ur}^- \text{ reabsorption} \rightarrow \downarrow \text{Ur}^- \text{ excretion} \rightarrow [\text{Ur}^-]_{\text{PI}}$
- Transcellular reabsorption:
 - Apical membrane:
 - Mediated by URAT1, OAT4 & OAT10
 - Basolateral membrane:
 - Mediated by URATv1 (voltage-driven).

Secretion

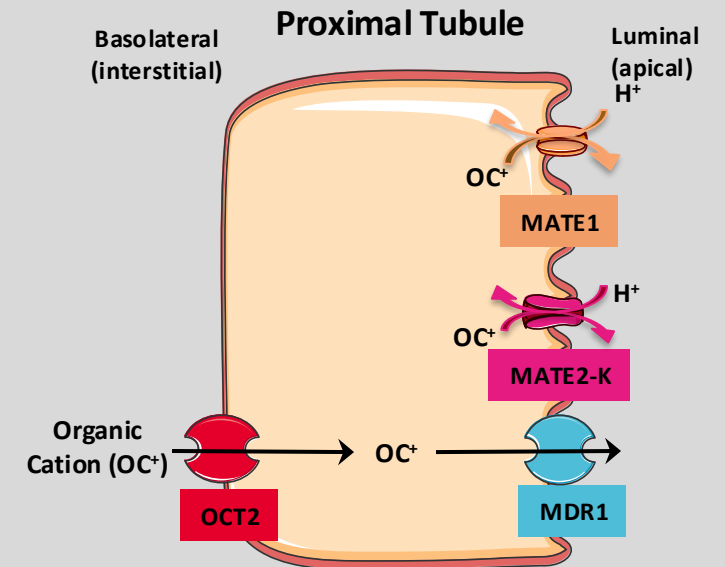
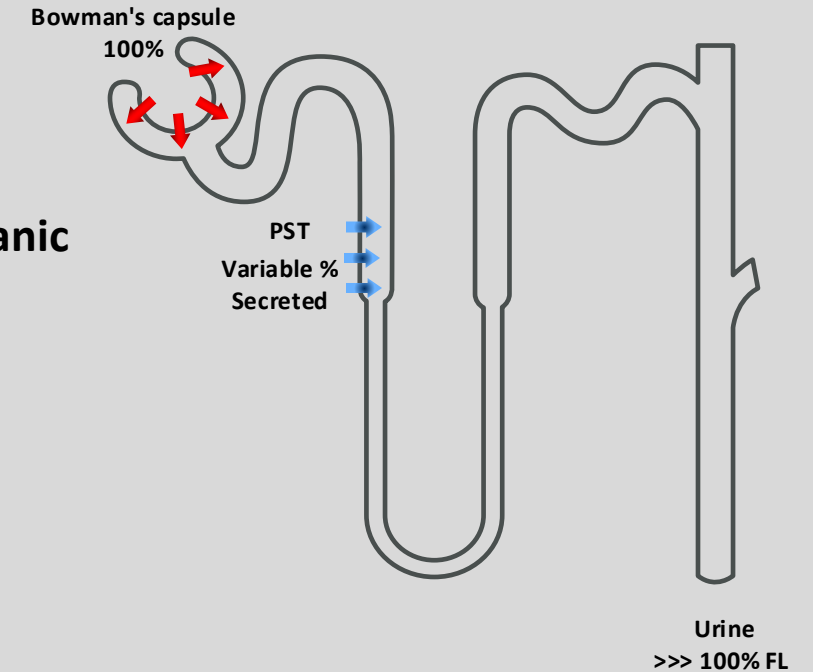
- Basolateral membrane: via OAT1 and OAT3
- Apical membrane: via NPT1/NPT4 & and two ABC transporters MRP4 & BCRP



Organic Base Secretory System (OASS)

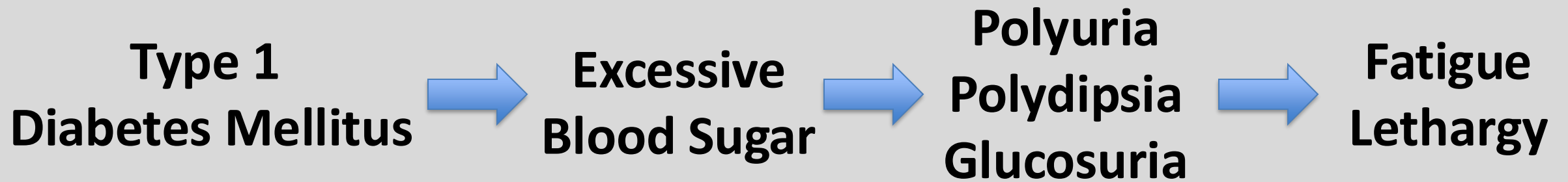
- PST (S3):
 - Responsible for secreting a wide range of endogenous and exogenous organic cations (organic bases).
- Important secreted endogenous organic cations:
 - Monoamine neurotransmitters (e.g., Dopa, Epi, NE, His)
 - Creatinine: the breakdown product of phosphocreatine.
- Important secreted exogenous organic cations:
 - Morphine, quinine, atropine.....
 - K^+ -sparing diuretics: Amiloride, triamterene.
- Driving force for basolateral membrane movement of OC^+ is the inside-negative membrane potential.
- Via a polyspecific organic cation transporter OCT2.
- At the apical membrane, exchangers MATE1, MATE2-K & MDR1 move OC^+ from cell to the lumen.
- Dependent on the H^+ gradient across the apical membrane.

Organic Base Reabsorption in the Nephron



Organic Anions & Cations Secreted by Proximal Tubule

	ANIONS	CATIONS
ENDOGENOUS	cAMP and cGMP	Dopa/Epi/NE/Ach/His/5HT
	Bile salts	Choline
	Oxalate/ α -KG	Thiamine
	Short-chain fatty acids (<6C)	Guanidine
	Prostaglandins	Creatinine
	Urate	
	Creatinine	
EXOGENOUS	Acetazolamide (CAI)	Amiloride & Triamterene (K ⁺ -Sparing)
	Chlorothiazide (Thiazide)	
	Furosemide (Loop)	
	Penicillin G	
	Probenecid	
	Saccharin	
	Salicylate (NSAID)	
	Indomethacin (NSAID)	
CONJUGATED	Glucuronate/Glutathione/Sulfate	



Lecture Feedback Form

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